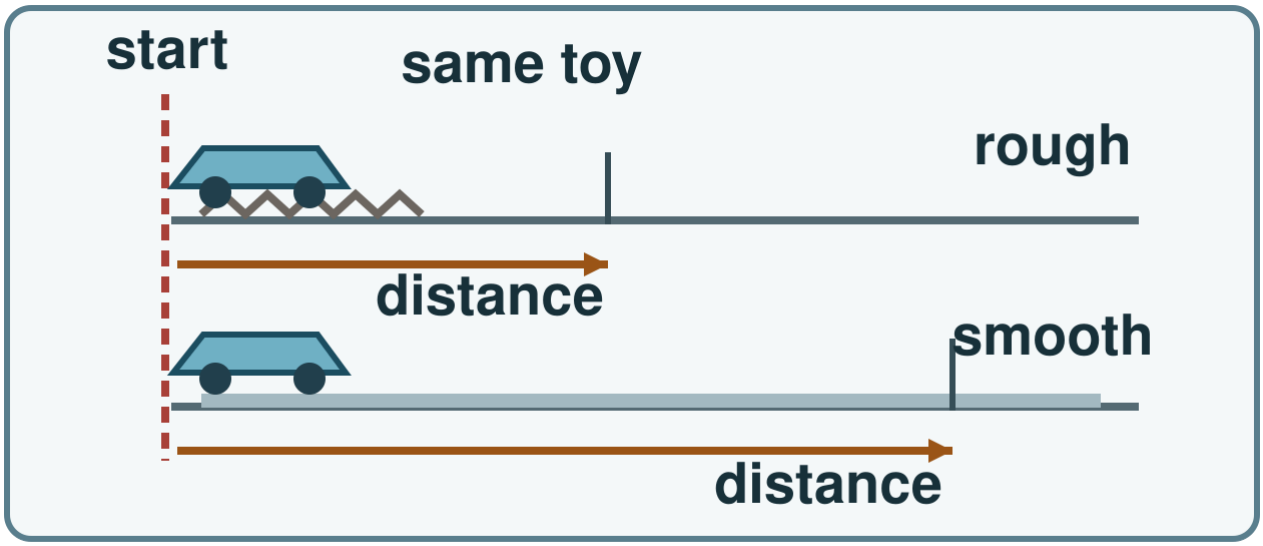
Knowledge organiser

I can name what I change and what I keep the same.

I can record what happened in a small test.

I can explain one way to make the test more dependable.

I am learning to… plan a fair investigation, record a result and explain a limit.



change only the surface

Use the same toy and start line. Change only the surface; measure distance. End marks are examples, not results.

I can name what I change and what I keep the same. I can record what happened in a small test. I can explain one way to make the test more dependable.

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Version _____

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Word page

I can name what I change and what I keep the same.

I can record what happened in a small test.

I can explain one way to make the test more dependable.

Word	It means	Example
variable	a factor that can change in a test	The surface is our changed variable.
fair test	a test that changes one key factor and keeps other key factors the same	We use the same toy in both runs for a fair test.
result	what you observe or measure in a test	Our result is the distance the toy travelled.
repeat	do the test again under the same conditions	We repeat each run to check our result.

Change one thing. Keep the toy, start and launch the same. Measure each run, keep each result and give a limit.

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Arrival and starter

I can name what I change and what I keep the same.

I can record what happened in a small test.

I can explain one way to make the test more dependable.

Choose the question card: does surface change how far this toy goes? Point to what we will change.

Think back. Say, point or write one answer for each question.

1. What should match a science claim?

2. Name an effect of added greenhouse gases.

3. What do we keep the same in a fair test?

You may point, sign, say or write your first ideas.

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I do · model and check

I can name what I change and what I keep the same.

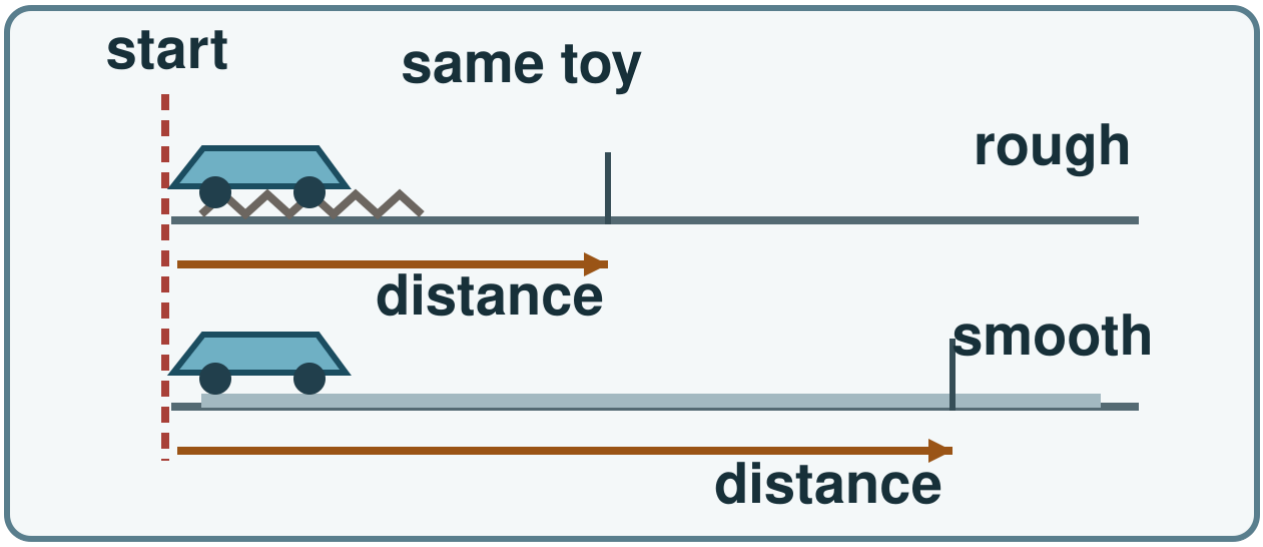
I can record what happened in a small test.

I can explain one way to make the test more dependable.

I mark a start line. I keep the toy and launch the same. I change the surface, measure distance and repeat. Runs may still vary.

Watch: Change rough/smooth surface → keep toy, start and launch the same → measure distance → repeat.

Show me: If I change the surface, should I also change the toy?



change only the surface

Use the same toy and start line. Change only the surface; measure distance. End marks are examples, not results.

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We do · sort and explain

I can name what I change and what I keep the same.

I can record what happened in a small test.

I can explain one way to make the test more dependable.

The plan has a fixed toy and start line. Add the missing launch control. Name the surface to change.

Our turn: Change ____ → keep toy and ____ the same → measure distance.

Talk, then sort.

Card	Choose a group
Surface type	change / keep same / measure
Toy	change / keep same / measure
Start line	change / keep same / measure
Travel distance	change / keep same / measure

Supported: Use a visual card for toy, start mark and launch; point to the missing control.

Independent: Name two controls and record a result.

Aspire: Explain how a different launch could make a surface comparison misleading.

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We do 2 · choose and explain

I can name what I change and what I keep the same.

I can record what happened in a small test.

I can explain one way to make the test more dependable.

What should you do if two repeat runs disagree?

A. Keep results and check the method

B. Alter one result to match

All of us: A team compares carpet and vinyl using one trolley. Which plan lets them test the surface fairly?

A. Same push; change start and surface.

B. Same start; change trolley and surface.

C. Same surface; change trolley and push.

D. Same start and push; change surface.

Show your choice and say why. You may use a point, card, board or sign. Take one quiet minute to reset.

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Independent · Supported

I can name what I change and what I keep the same.

I can record what happened in a small test.

I can explain one way to make the test more dependable.

Test two runs per surface or observe. Record distance. Name what changed, stayed the same and could limit the test.

Talk first. You can plan in any language.

Point, sign, speak, draw, type, write, use a scribe or show a staff-captured photo.

Point to the factor that changes.

Say first: I change the ___; I keep ___ the same.

Surface	Run 1 (cm)	Run 2 (cm)
Rough	_____	_____
Smooth	_____	_____

Choose one route within the full 8 minutes. Keep your own response.

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Independent · Independent

I can name what I change and what I keep the same.

I can record what happened in a small test.

I can explain one way to make the test more dependable.

Test two runs per surface or observe. Record distance. Name what changed, stayed the same and could limit the test.

Talk first. You can plan in any language.

Point, sign, speak, draw, type, write, use a scribe or show a staff-captured photo.

Name two controls and record a result.

Say first: I changed ___ and kept ___ the same. My result was ___.

Surface	Run 1 (cm)	Run 2 (cm)
Rough	_____	_____
Smooth	_____	_____

Choose one route within the full 8 minutes. Keep your own response.

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Independent · Aspire

I can name what I change and what I keep the same.

I can record what happened in a small test.

I can explain one way to make the test more dependable.

Test two runs per surface or observe. Record distance. Name what changed, stayed the same and could limit the test.

Talk first. You can plan in any language.

Point, sign, speak, draw, type, write, use a scribe or show a staff-captured photo.

Explain why repeat runs help and identify a method limit.

Say first: My result suggests ___, but I would repeat ___ because ___.

Go deeper: Two runs differ. Name a reason and fix a method step.

Fair test repair: Two repeats differ. Name a safe conclusion, then improve the method for the next test.

Surface	Run 1 (cm)	Run 2 (cm)
Rough	_____	_____
Smooth	_____	_____

Choose one route within the full 8 minutes. Keep your own response.

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Practical model · safe steps and record

I can name what I change and what I keep the same.

I can record what happened in a small test.

I can explain one way to make the test more dependable.

Guess first: Will the toy go farther on rough or smooth paper? Give a reason.

Kit: Small toy · Low launch guide · Rough and smooth paper · Start-line tape · Ruler

Mark one start and use the same toy.

Launch twice on rough paper in the same way.

Measure and record each distance.

Repeat on smooth paper; change only the surface.

Compare runs and name a limit.

Risk: A launched toy may hit someone or drop off the track.

Safe: Keep the track low and clear. Retrieve the toy only after it stops.

Surface	run 1 (cm)	run 2 (cm)
Rough	___	___
Smooth	___	___

Use the kit your teacher has checked. A picture or adult-led model is a valid route.

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Review and exit

I can name what I change and what I keep the same.

I can record what happened in a small test.

I can explain one way to make the test more dependable.

Check the controls and result with a partner. Fix one label before you conclude.

VF: hear one comment. NS(+): name a next step. //: edit one part. ?: ask for a read-back. Take one quiet minute.

Name the change, one control, one result and a next step.

? Read it back: plan a fair investigation, record a result and explain a limit.

My feedback was… say what changed. Next I need to… name a fix.

How I know: I can name what I change and what I keep the same. I can record what happened in a small test. I can explain one way to make the test more dependable.

The adult hears you and chooses our next example. Take one quiet minute.

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